

Problem Statement 1:

Nowadays, maintaining attention during online classes and computer-based learning is difficult for many students. Traditional methods of checking attention are manual, time-consuming, and not suitable for real-time monitoring. To solve this problem, this project proposes a real-time attention detection system using eye blink behavior and head movement analysis through a webcam. The system automatically detects user concentration levels and classifies attention as high, medium, and low.

What I Did in This Project

In this project, I developed a real-time attention monitoring system using computer vision, machine learning, and deep learning techniques. The system detects eye blinks, blink duration, blink frequency, Eye Aspect Ratio (EAR), and head movements such as pitch, yaw, and roll. Using these features, the system calculates attention percentage and concentration level continuously from webcam video frames.

Type of Landmark Used

The system uses facial landmarks to detect important points around the eyes and face. A MediaPipe landmark model was used to identify left-eye and right-eye regions for EAR calculation using 7 vertical and 1 horizontal and blink detection. The eye landmarks help determine whether the eyes are open or closed, while facial landmarks around the nose and face are used for head pose estimation and movement tracking.

Features - Attention Detection System

- | — Eye Blink Features
 - | | — Left EAR, Right EAR, Average EAR, Dynamic Threshold, Ear smoothed
 - | | — Blink Detection (Yes, No), Blink Duration, Blink Stage (No, Blink starts, Blink continues, Blink ends)
 - | | — Blink Count, Short Blink Count (per min), Normal Blink Count (per min), Long Blink Count (per min)
 - | | — Blink Type Fuzzy Approach A, Blink Type Fuzzy Approach B
- | — Eye Status Features - Eye Status (Active - Open, Inactive - Close)
- | — Attention Features - Attention Percentage, Attention Level
- | — Head Movement Features
 - | | — Pitch, Yaw, Roll, Head Movement, Head Level
- | — Time Features - Frame Number, Minute

Model Training Strategy

Initially, attention level and head level features were considered as target labels for machine learning and deep learning model training. However, most of the collected samples belonged to the "Active" and "High" attention categories and nearly 100% attention, which caused class imbalance and overfitting problems during training. Due to this issue, these labels were not suitable for reliable model prediction and generalization.

To overcome this limitation, quiz performance scores were used as the final classification labels instead of directly using attention percentage or head-level values. Based on quiz marks, students were categorized into high, medium, and low-performance groups. The extracted eye blink features and head movement features were then used as input features for training the machine learning and deep learning models. This approach helped create a more realistic relationship between visual attention behavior and actual student performance while reducing overfitting problems.

Problem Statement 2

Traditional attention monitoring systems mainly rely on eye blink detection and head movement analysis to estimate student concentration levels during online learning. However, these methods alone may not accurately identify whether the student is truly focusing on the learning content. A student may keep their eyes open and maintain normal head posture while still looking away from the screen or becoming distracted. Therefore, there is a need for a more intelligent and reliable attention monitoring system that can additionally analyze gaze direction and emotional behavior.

To address this problem, the proposed system aims to integrate gaze detection and emotion recognition with existing eye blink and head movement analysis for real-time attention monitoring. Gaze detection can identify whether the student is looking at the screen, looking away, or becoming distracted by tracking eye direction and pupil movement. Emotion detection can further analyze facial expressions such as boredom, confusion, tiredness, stress, or interest during learning sessions. By combining eye blink features, gaze direction, head pose estimation, and emotional states, the system can provide a more accurate and robust estimation of student attention levels in online learning environments.

Gaze Detection for Attention Monitoring

In the proposed system, gaze detection can be implemented using MediaPipe Face Mesh landmarks. MediaPipe provides detailed eye landmarks that help track pupil position and eye direction in real time through a webcam. The system first detects facial landmarks and extracts eye region points from both left and right eyes. Using these landmarks, the pupil center position is calculated relative to the eye corners. Based on the pupil location, the gaze direction can be classified into left, right, up, down, and center.

For example, if the pupil moves toward the left side of the eye region, the gaze is classified as “looking left.” Similarly, movement toward the right, upper, or lower regions indicates different gaze directions. When the pupil remains near the center of the eye region, the gaze is considered focused on the screen. These gaze directions can then be converted into attention scores. Looking at the center may produce high attention scores, while continuously looking away from the screen may reduce the attention score. This approach helps identify distraction levels more accurately than blink analysis alone.

Gaze Direction → Attention Score

Center → High Attention

Left → Medium Attention

Right → Medium Attention

Up → Low Attention

Down → Low Attention

Frequent Direction Change → Distracted

Emotion Detection for Attention Monitoring

In the proposed system, emotion detection can be implemented using facial expression analysis with MediaPipe Face Mesh and deep learning models such as CNN. The webcam continuously captures the student's face, and facial landmarks are extracted from regions such as the eyes, eyebrows, nose, and mouth. Based on facial muscle movements and expressions, the system classifies emotions into seven standard categories: Happy, Sad, Angry, Fear, Surprise, Disgust, and Neutral.

These seven emotions can be used to estimate the student's engagement and attention level during online learning sessions. For example, Neutral and Happy emotions may indicate active participation and concentration, while Sad or Fear emotions may represent confusion or reduced engagement. By continuously monitoring emotional changes, the system can better understand student learning behavior and improve real-time attention monitoring.

Emotion → Attention Interpretation

├— Happy → High Attention

├— Neutral → Focused State

├— Surprise → Active Response

├— Sad → Low Engagement

├— Angry → Disturbed Attention

├— Fear → Uncomfortable State

└─ Disgust → Distraction

Blink Frequency

Blink frequency means the number of eye blinks occurring within a specific time period, usually one minute. It helps identify concentration and fatigue levels. Very low blink frequency may indicate high focus, while very high blink frequency may indicate tiredness, stress, or distraction.

Blink Frequency = Total Number of Blinks / Time (minutes)

Example:

Blink Frequency = 20 blinks / 1 minute = 20 BPM

(BPM = Blinks Per Minute)

Blink Attention Score Calculation

The blink attention score is calculated using EAR, blink duration, and blink frequency. Normal blinking behavior gives higher attention scores, while long eye closures reduce attention scores.

Blink Attention Score = $(0.4 \times \text{EAR Score}) + (0.3 \times \text{Blink Duration Score}) +$
 $(0.3 \times \text{Blink Frequency Score})$

Example Rules

└─ Normal Blink Frequency → High Score

└─ Long Eye Closure → Low Score

└─ Short Natural Blink → Medium/High Score

└─ Excessive Blinking → Low Attention

Gaze Attention Score Calculation

The gaze attention score is calculated based on eye direction. Looking at the screen center increases attention score, while continuously looking away decreases the score.

$$\text{Gaze Attention Score} = \text{Center Gaze Frames} / \text{Total Frames} \times 100$$

Example:

If Center Frames = 850

Total Frames = 1000

$$\text{Gaze Score} = (850 / 1000) \times 100 = 85\%$$

Gaze Direction Scores

└─ Center → 100

└─ Left → 60

└─ Right → 60

└─ Up → 40

└─ Down → 30

Emotion Attention Score Calculation

The emotion attention score is calculated using detected facial emotions. Positive and neutral emotions increase the attention score, while tired or negative emotions reduce the score.

$$\text{Emotion Attention Score} = \text{Emotion Weight} / \text{Maximum Weight} \times 100$$

Emotion Weights

└─ Neutral → 100

└─ Happy → 90

└─ Surprise → 80

└─ Fear → 50

|— Sad → 40

|— Angry → 35

└— Disgust → 30

Head Attention Score Calculation

The head attention score is calculated using pitch, yaw, and roll angles. Looking forward continuously gives a higher score, while excessive head movement lowers attention.

Head Attention Score = Forward-Looking Frames / Total Frames $\times$ 100

Example:

If Forward Frames = 920

Total Frames = 1000

Head Score = $(920 / 1000) \times 100 = 92\%$

Head Movement Rules

|— Looking Forward → High Score

|— Looking Left/Right → Medium Score

|— Frequent Movement → Low Score

└— Head Down → Very Low Score

Final Attention Score Formula

The final attention score combines all module scores using weighted fusion.

Final Attention Score = $(0.30 \times \text{Blink Score}) + (0.35 \times \text{Gaze Score}) + (0.20 \times \text{Emotion Score}) + (0.15 \times \text{Head Score})$

Final Attention Classification

|— 80 – 100 → High Attention

|— 50 – 79 → Medium Attention

└─ 0 – 49 → Low Attention

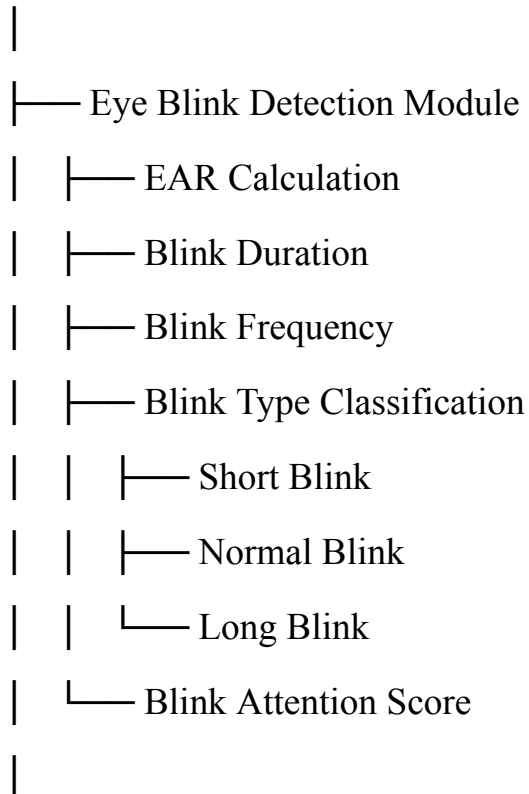
Combined Attention Framework

In the future, the system can combine eye blink detection, gaze tracking, head movement analysis, and emotion recognition into a single multimodal attention monitoring framework. The combined features can improve the accuracy and robustness of attention prediction compared to using only blink-based features. This integrated approach can better represent real student concentration levels during online education environments.

Future Deep Learning Enhancement

Future work can also include advanced deep learning models such as CNN-LSTM, Transformers, or multimodal neural networks to analyze temporal behavioral patterns. These models can learn relationships between eye movement, gaze direction, emotions, and quiz performance over time. The final system can be used for smart classrooms, e-learning platforms, examination monitoring, and adaptive learning systems.

Attention Monitoring System



└─ Gaze Detection Module

| └─ Left

| └─ Right

| └─ Up

| └─ Down

| └─ Center

| └─ Gaze Attention Score

|

└─ Emotion Detection Module

| └─ Happy

| └─ Sad

| └─ Angry

| └─ Fear

| └─ Surprise

| └─ Disgust

| └─ Neutral

| └─ Emotion Attention Score

|

└─ Head Movement Module

| └─ Pitch

| └─ Yaw

| └─ Roll

| └─ Looking Forward

